Frequency Lock Encoding

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***Abstract*—Modern wireless systems are designed with excess synchronization bandwidth to ensure reliable operation under worst-case conditions. This paper presents a protocol-agnostic secondary signaling layer that exploits this unused synchronization margin by intentionally introducing small, controlled frequency offsets into an existing communication signal to embed a parallel stream of information. These offsets are naturally absorbed by the primary receiver's carrier frequency offset (CFO) correction mechanisms, allowing the primary data stream to remain intact while enabling simultaneous secondary communication. The proposed signaling method employs a frequency-shift-keyed overlay engineered to remain within the stable operating region of conventional carrier synchronization loops, ensuring backward compatibility and transparency to legacy receivers. We analyze the system for single-carrier protocols, characterize the trade-offs between secondary throughput and impact on the primary link, and validate performance through both simulation and over-the-air experiments using software-defined radios. Results demonstrate reliable secondary communication with negligible degradation to primary performance, making our system well-suited for ad hoc spectrum sharing and decentralized coordination at the tactical edge.**

***Index Terms*—tactical communications, spectrum sharing and coordination, subprotocol, phase-locked loop, software-defined radio**

I. Introduction

Wireless links face unpredictable analog disturbances that introduce errors into transmitted signals. To ensure reliability under worst-case conditions, protocols are designed with powerful synchronization mechanisms that reserve additional bandwidth to tolerate impairments. Because failure is often catastrophic, designers consistently provision systems for worst-case operating conditions. As a result, modern wireless systems contain a hidden inefficiency: *excess impairment-correction capacity* under many operating environments. While this safety margin guarantees robustness, it also means that valuable bandwidth is frequently spent protecting against errors that are not actually present.

In this paper, we propose a different perspective on signal impairments and noise in wireless communication. Instead of treating unused synchronization capacity as wasted overhead, we intentionally and opportunistically introduce *controlled, meaningful impairments* into a signal to create a secondary layer of communication without disrupting the primary communication. To the primary protocol, these perturbations appear as ordinary noise; however, to a receiver that knows how to interpret them, they represent a *second parallel channel of information*. This secondary channel can support a fully featured "subprotocol" capable of carrying coordination data, such as in-band signaling between unassociated networks without prior coordination or spectrum usage information for agile spectrum operations. In this way, what would otherwise be unused correction capacity becomes a new signaling resource that enables simultaneous communication *without requiring additional spectrum, antennas, or modifications to the underlying protocol*.

This approach is powerful for three reasons. First, it is protocol-agnostic. Because the subprotocol operates by embedding information within controlled perturbations that are corrected by the primary receiver's synchronization algorithms, it does not depend on the modulation, framing, or higher-layer structure of the underlying protocol. Second, it is backward compatible and, as a result, can be used with existing commercial technologies. A legacy receiver that is unaware of the subprotocol simply interprets the intentional perturbations as ordinary noise. Third, it is opportunistic. The subprotocol consumes only the excess impairment-correction capacity available in the primary link. When the full synchronization capability of the primary system is required, the secondary signaling can be disabled entirely without affecting normal operation. These properties make the proposed approach well-suited for decentralized spectrum coordination and resilient edge communications.

Although many forms of meaningful impairments could be used to realize such a subprotocol, this work focuses specifically on embedding information through controlled frequency offsets applied to the underlying communication waveform, as illustrated in Fig. 1. The choice of frequency perturbations over other synchronization domains, such as timing, is deliberate. Frequency correction feedback loops generally do not suffer from self-noise in the same manner as timing recovery loops, resulting in a cleaner observable signal. Additionally, while frequency synchronization is often implemented in hardware, the outputs of these algorithms are commonly exposed to firmware or software layers, unlike many synchronization mechanisms in other domains.

Others have used similar techniques to provide covert communication [1]–[4], which we will discuss in the next section. However, the primary goal of this paper is to establish

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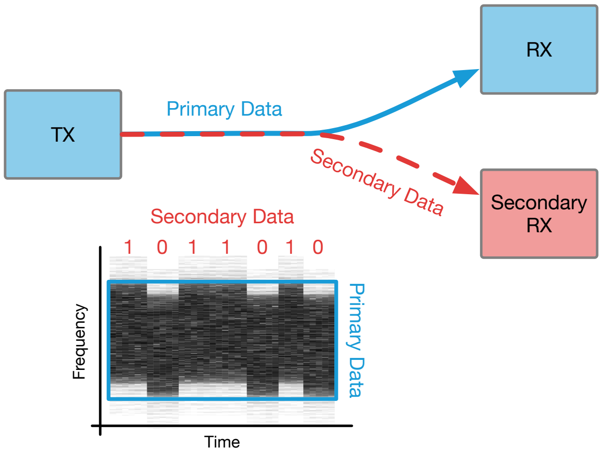

Fig. 1. By intentionally introducing a controlled frequency offset into the primary signal, secondary spectrum coordination information can be conveyed simultaneously.

the mathematical foundation for such a subprotocol and to demonstrate its practical feasibility for spectrum coordination through analysis, preliminary simulations, and software-defined radio (SDR) experiments. To this end, we make the following contributions:

- We introduce the concept of encoding a secondary stream of information into a primary transmission through controlled variation of the carrier frequency offset (CFO), which we call **Frequency Lock Encoding** (FLE). Although the method is protocol-agnostic, we develop and evaluate it in the context of single-carrier waveforms.
- We develop the mathematical framework necessary to describe and analyze FLE, deriving closed-form expressions for the signal-to-noise ratio (SNR) available to the secondary detector and for the range of frequency deviations and secondary symbol rates over which the primary carrier loop remains locked.
- We develop a high-fidelity simulator for FLE over a single-carrier protocol in GNU Radio Companion and demonstrate that, under the conditions predicted by our analysis, FLE introduces little to no measurable degradation to the primary modulation scheme.
- We implement a proof-of-concept realization using SDRs and perform preliminary over-the-air experiments demonstrating that the approach is practically viable while achieving data rates of over 900 kbps.

II. Related Work

A. Covert Communication

The idea of leveraging waveform parameters to embed covert data has precedent [1]–[4]. Such approaches range from embedding data in constellations by emulating noisy channel conditions [1] to emulating hardware noise using machine learning [2]. The previous work most similar to our work was done by Classen et al. [4], where the authors explore multiple covert channels for Wi-Fi systems, one of which applies a static carrier frequency offset to each Wi-Fi frame and recovers it from the preamble frequency estimate. Their offsets span 1–50 kHz and are selected empirically per platform: below roughly 5 kHz the covert channel becomes unreliable, and above roughly 10 kHz the legitimate frame error rate begins to rise. Because the offset is held constant over a frame and read from a per-frame estimator, the achievable rate is tied to the frame rate rather than to the symbol rate, yielding 250 kbps for 802.11a/g.

Our work differs in the mechanism, not merely in the waveform it is applied to. FLE modulates a *continuously varying*, pulse-shaped offset that the primary receiver's carrier tracking loop absorbs by design, so the secondary symbol rate is referenced to the primary *symbol* rate rather than to a framing boundary. This is what allows the technique to operate on continuously transmitted single-carrier waveforms, which expose no per-frame frequency estimate to reuse, and it is the regime in which most tactical waveforms operate. It also changes how the operating point is chosen: rather than tuning the offset empirically on a specific platform, Section III-B derives the admissible range of frequency deviation and secondary symbol rate in closed form from the primary loop's bandwidth and damping, so the design transfers to any receiver whose carrier synchronizer is characterized. Finally, while FLE could be used for covert communication, we focus on coordination rather than covertness, and as a result the goals of our system significantly differ: we are more concerned with protocol coexistence, both in the context of spectrum coordination and between the primary link and the subprotocol. We have also developed a fully featured and usable subprotocol, with its own framing and error detection, whereas others have focused on avoiding detection.

B. Spectrum Coordination

Large-scale spectrum coordination is often performed using a centralized database, such as that used by CBRS [5] and Wi-Fi 6E [6]. However, single points of failure and dependence on backhaul connectivity are poorly suited to tactical operations in denied or disconnected environments. Our approach enables peer-to-peer spectrum coordination without requiring a centralized database.

Other ad hoc spectrum-sharing protocols have been explored. Similar to our approach, they modify transmission characteristics to embed spectrum-sharing data. These methods include embedding spectrum data in an OFDM subcarrier [7], in the amplitude of a signal [8], in a sequence of transmissions using on-off keying [9], and in inter-transmission timing [10]. While similar in goal, our approach differs substantially from these works. We use frequency offsets to encode secondary data, yielding a much higher data rate (hundreds of kbps compared to tens of bps).

III. System Design

In developing FLE, we adhere to the following two design constraints: avoid degrading the primary communication while sending as much data as possible through the secondary channel. To this end, we explore the mathematical bounds of such a system in the context of a single-carrier system and describe how it can be used for spectrum coordination.

A. Overview

The signals generated by the primary and secondary systems are illustrated in Fig. 2. The complex-valued lowpass equivalent [11] of the primary signal is

$$s_1(t) = \sum_k a_{1,k} p_1(t - kT_{s,1}) \tag{1}$$

where $a_{1,k}$ is the k-th QAM symbol (or constellation point), $p_1(t)$ is the pulse shape, and $T_{s,1}$ is the symbol time. The secondary signal begins with a pulse amplitude modulation (PAM) pulse train

$$m(t) = \sum_{k'} a_{2,k'} p_2(t - k'T_{s,2}) \tag{2}$$

where $a_{2,k'}$ is the k'-th real-valued PAM symbol, $p_2(t)$ is a pulse shape normalized so that its $T_{s,1}$-spaced samples have unit ℓ_2 norm, and $T_{s,2}$ is the symbol time. The PAM pulse train $m(t)$ is frequency modulated to produce the complex-valued lowpass equivalent signal $s_2(t) = e^{j\phi(t)}$ where

$$\phi(t) = 2\pi\Delta f \int_0^t m(u)du \tag{3}$$

$$= 2\pi\Delta f \sum_{k'} a_{2,k'} q_2(t - k'T_{s,2}) \tag{4}$$

$$q_2(t) = \int_0^t p_2(u)du \tag{5}$$

The complex-valued lowpass equivalent of the transmitted signal using FLE is

$$s(t) = s_1(t)e^{j\phi(t)} \tag{6}$$

The phase variations in $s(t)$ are due to the secondary transmission. The modulation index Δf should be small enough to allow the PLL at the detector to track out the phase variations.

The received signal is

$$r(t) = s_1(t)e^{j\theta(t)} + w(t) \tag{7}$$

where

$$\theta(t) = \vartheta(t) + \phi(t) \tag{8}$$

where $\vartheta(t)$ is the phase shift due to the propagation medium and $w(t)$ is the thermal noise modeled as a white wide-sense stationary complex-valued normal random process with zero mean and power spectral density $2N_0$ W/Hz [11]. A block diagram of the channel is illustrated in Fig. 2(a).

The detectors for the primary and secondary data are shown in Fig. 2(b). The received signal is applied to a filter matched to the primary pulse shape $p_1(t)$ and sampled every $T_{s,1}$ seconds in synchronism with the symbol transitions on the primary data channel. (A symbol timing synchronizer is assumed, but not shown.) The matched filter outputs form a discrete-time sequence that is applied to a decision-directed phase-locked loop (PLL). The k-th matched filter output $z(kT_{s,1})$ is derotated by the k-th phase estimate $\hat{\theta}(kT_{s,1})$ to produce $z_r(kT_{s,1})$. The phase error detector (PED) produces the k-th error signal $e(kT_{s,1})$ based on the maximum likelihood principle [12]. The error signal $e(kT_{s,1})$ is applied to a loop filter with z-domain transfer function $F(z)$ to produce $v(kT_{s,1})$. The loop filter output is applied to a direct digital synthesizer (DDS) to produce the phase estimate used for derotating the next matched filter output.

When in lock, the loop filter output $v(kT_{s,1})$ is a noisy version of $T_{s,1}$-spaced samples of the derivative of $\theta(t)$. To see this, assume $p_1(t)$ satisfies the Nyquist no-ISI condition, so that the k-th matched filter output is $z(kT_{s,1}) = a_{1,k}e^{j\theta(kT_{s,1})} + \nu(kT_{s,1})$, where ν is a complex-valued white Gaussian sequence of variance $2N_0$. When the phase error $\theta_e = \theta - \hat{\theta}$ is small enough that the primary symbol decisions are correct, the maximum likelihood PED output reduces to

$$e(kT_{s,1}) = E_{avg} \sin\theta_e(kT_{s,1}) + \nu_0(kT_{s,1}) \tag{9}$$

where E_{avg} is the average primary symbol energy and ν_0 is a real-valued white Gaussian sequence of variance $E_{avg}N_0$. The PED gain is therefore $K_p = E_{avg}$. Referring the noise to the loop input as $\nu_1 = \nu_0/K_p$, the linearized loop maps its input $\theta(kT_{s,1}) + \nu_1(kT_{s,1})$ to the loop filter output $v(kT_{s,1})$ and to the phase estimate $\hat{\theta}(kT_{s,1})$ through

$$H_2(z) = \frac{K_p\left(1 - z^{-1}\right)F(z)}{1 - z^{-1}\left[1 - K_pF(z)\right]} \tag{10}$$

$$H_3(z) = \frac{z^{-1}}{1 - z^{-1}} H_2(z) \tag{11}$$

respectively. Because N is large and Δf is small, $\theta(t)$ is heavily oversampled at the primary symbol rate, so only $\Omega \approx 0$ is of interest. Substituting the first-order approximation $e^{-j\Omega} \approx 1 - j\Omega$ into (10) gives

$$H_2(e^{j\Omega}) \approx j\Omega, \qquad \Omega \approx 0 \tag{12}$$

The loop filter therefore differentiates its input phase, and differentiating (8) gives

$$v(kT_{s,1}) \approx \vartheta'(kT_{s,1}) + 2\pi\Delta f \sum_{k'} a_{2,k'} p_2(kT_{s,1} - k'T_{s,2}) + \text{noise} \tag{13}$$

The approximation (12), and with it (13), holds only when the bandwidth of the secondary pulse train is less than the closed-loop bandwidth of the PLL. The following three scenarios present no problem to detecting the secondary signal:

1) If the propagation medium induces a static phase shift, then $\vartheta(t) = \vartheta$ whose time derivative is zero.
2) If the propagation medium induces a static frequency offset, then $\vartheta(t)$ is the phase ramp $\vartheta(t) = \Delta\omega t$ whose time derivative is the constant $\Delta\omega$. This constant may be removed from the sample sequence $v(kT_{s,1})$ using a DC blocker.
3) If the propagation medium induces a slowly varying phase shift that can be considered quasi-static, then the derivative of $\vartheta(t)$ is small relative to the secondary pulse train.

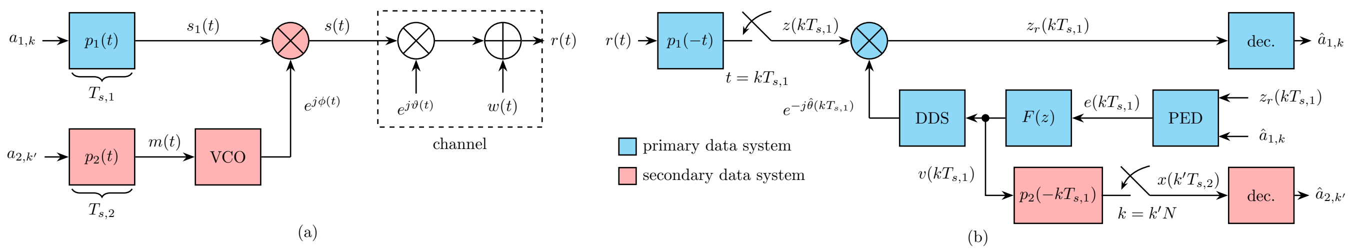

Fig. 2. A block diagram illustrating the FLE technique for single-carrier modulation: (a) the transmitted signal $s(t)$ and the received signal $r(t)$; (b) the FLE detector (the loop filter output of the primary data serves as the input for secondary data detection).

Under any one of them, $v(kT_{s,1})$ carries the secondary PAM pulse train alone, so detection may be accomplished by applying $v(kT_{s,1})$ to a discrete-time FIR filter matched to $T_{s,1}$-spaced samples of the secondary pulse shape. Every $N = T_{s,2}/T_{s,1}$-th sample of the matched filter output is used for detection of the secondary symbols. The optimum phasing of the downsampling operation is controlled by a symbol-timing synchronizer (not shown).

Cases 2 and 3 above are what platform mobility and oscillator drift reduce to in practice, and they show that the subprotocol is degraded not by Doppler itself but by the *rate of change* of Doppler—and then only when that rate approaches the secondary symbol rate $1/T_{s,2}$. The separation between the two is large. For a platform closing at 300 m/s with a 1 km miss distance at a 915 MHz carrier, the Doppler shift sweeps at roughly $v^2/\lambda d \approx 270$ Hz/s, so it changes by a few millihertz over one secondary symbol, an impairment that is negligible during secondary signal detection. The channel term $\vartheta'(kT_{s,1})$ in (13) is thus orders of magnitude below the subprotocol term for realistic tactical geometries, and the residual constant is removed by the DC blocker in any case. The regime that does defeat this argument is high mobility combined with fast frequency-selective fading, in which $\vartheta(t)$ ceases to be quasi-static; that case is the principal limitation of the technique.

B. Design Constraints and Parameters

The derivation above leaves two free parameters and each is bounded from both directions to ensure primary and subprotocol signal reception. Throughout we take the secondary alphabet to be binary, $a_{2,k'} \in \{-A_2, +A_2\}$ with $A_2 = 1$ in all numerical results, and assume the primary link uses a second-order PLL with the proportional-plus-integrator loop filter $F(z) = K_1 + K_2/(1 - z^{-1})$ [12], damping factor ζ, and closed-loop equivalent noise bandwidth $B_nT_{s,1}$.

The modulation index Δf, or its normalized version $\Delta fT_{s,1}$, controls the magnitude of the frequency deviation induced by the secondary PAM pulse train. Its *lower* bound is set by noise. The two links observe the same thermal noise through different transfer functions: passing ν_1 through H_3 gives the phase jitter seen by the primary detector,

$$\sigma_{\hat{\theta}}^2 = \frac{2B_nT_{s,1}}{E_{avg}/N_0} \tag{14}$$

with $B_nT_{s,1} = \frac{1}{4\pi}\int_{-\pi}^{\pi} |H_3(e^{j\Omega})|^2 d\Omega$ the closed-loop equivalent noise bandwidth [12], [13], while passing the same noise through H_2 gives the noise variance at the input to the secondary detector, $\sigma_v^2 = I_2 E_{avg} N_0 / K_p^2$, where $I_2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |H_2(e^{j\Omega})|^2 d\Omega$. Since the secondary matched filter output is $2\pi\Delta fT_{s,1}a_{2,k'}$ plus this noise, the ratio governing the secondary probability of error is

$$\frac{E_{avg,2}}{\sigma_v^2} = \frac{(2\pi\Delta fT_{s,1}A_2)^2}{I_2}\,\frac{E_{avg}}{N_0} \tag{15}$$

Reducing $\Delta fT_{s,1}$ too far therefore drops the secondary signal into this noise floor. Note also that the proportionality to E_{avg}/N_0 in (15) is exact, so the two links degrade together: the secondary channel inherits the operating range of the primary channel rather than possessing an independent one.

The *upper* bound on Δf is set by the requirement that the primary PLL remain locked, since a deviation the loop cannot track out compromises the primary signal. The secondary pulse train is an FM modulating signal, so Gardner's modulation limits for a second-order PLL apply [14]. Two bounding cases are informative. An alternating symbol sequence produces the fastest approximately sinusoidal deviation the subprotocol can generate, with peak deviation $\Delta fA_2\kappa/\sqrt{T_{s,2}}$, where $\kappa \approx 0.1414$ is a pulse-shape peak factor nearly independent of the excess bandwidth β of $p_2(t)$. An i.i.d. symbol sequence is better modeled as a bandlimited Gaussian process of bandwidth $(1+\beta)/2T_{s,2}$ and RMS deviation $\Delta fA_2/\sqrt{T_{s,2}}$. Applying [14, Ch. 5] to each case and normalizing to the primary symbol time gives the common form

$$\frac{\Delta fA_2T_{s,1}}{(B_nT_{s,1})^2 N}\cdot\frac{c}{\sqrt{T_{s,2}}} < \frac{C}{\pi^2\left(\zeta + \frac{1}{4\zeta}\right)^2}, \qquad N \gg \frac{\pi}{B_nT_{s,1}}\left(\zeta + \frac{1}{4\zeta}\right) \tag{16}$$

with $(c, C) = (\kappa, 1)$ for the sinusoidal case and $(c, C) = \left(1, \frac{2\sqrt{3}}{\gamma(1+\beta)}\right)$ for the random case, where $\gamma \approx 3.5$ is the empirical crest factor Gardner fits to laboratory observations of modulation unlock [14]. Since $\gamma(1+\beta) > 1$ in this application, the random-modulation limit is the more conservative of the two.

The secondary data rate $1/T_{s,2}$ matters relative to the primary data rate, a relationship represented by the ratio

$$N = \frac{T_{s,2}}{T_{s,1}} \quad \text{primary symbols per secondary symbol.} \tag{17}$$

The throughput of the secondary channel increases with decreasing N, but two effects bound N from below. First, N appears in the denominator of the bound (16), so shrinking it tightens the admissible deviation, working directly against the lower bound imposed by (15). Second, once N is small enough to violate the bandwidth condition on (12), both links suffer, though at different points. The loop no longer tracks out the phase variations, and (13) no longer describes $v(kT_{s,1})$.

Taken together, (15) and (16) provide several insights about the resulting operating envelope. First, the maximum admissible normalized deviation grows *linearly* in N. Second, it grows as $(B_nT_{s,1})^2$: a receiver with a wider carrier tracking loop—the usual choice for robust operation under Doppler and oscillator uncertainty—admits quadratically more secondary capacity, so the margin FLE consumes is not a fixed quantity but one that grows with the primary link's own robustness. Third, (16) provides a set of bounds for the empirical operating envelope of FLE. The pulse-shaped binary PAM train used here is less random than noise and more random than a sinusoid, so the true operating envelope is expected to fall between the two. This is shown in Section V-A.

Finally, it is reasonable to ask why the same margin could not simply be reclaimed by narrowing the synchronization loop instead. Equation (14) shows what that would buy: reducing $B_nT_{s,1}$ lowers the primary phase jitter, which yields a marginal bit error rate improvement that can be converted into throughput only by renegotiating modulation and coding with an *associated* peer, and at the cost of longer acquisition and a smaller pull-in range. FLE converts the same margin into bits legible to any receiver capable of estimating CFO, whether or not it is associated, keyed, or able to demodulate the payload. The two uses of the margin are therefore not interchangeable.

C. System Feasibility

While direct access to raw IQ samples or internal PLL signals is typically unavailable in commercial off-the-shelf (COTS) hardware (though not unheard of [15]), this work explores the capabilities that could be realized if such signals were more readily exposed at the hardware level. These signals already exist within the digital logic of modern radios; in many cases, only register access would need to be propagated through the software stack. Similarly, applying an FSK overlay via carrier mixing is generally infeasible in software alone, but modest hardware extensions could enable this functionality on most COTS designs. A key insight is that although minor architectural changes would be required, they would remain **fully backward compatible with existing communication protocols**. The deployment burden is also asymmetric: only the transmitter requires the carrier-mixing extension, whereas an FLE receiver can in many cases be realized without hardware changes at all, since most contemporary systems already expose packet-level CFO estimates that can be repurposed to recover secondary symbols.

D. Spectrum Coordination

This system is particularly well-suited for spectrum coordination where radios from different units may operate in close proximity without shared network association or reliable access to centralized spectrum-management infrastructure. Because FLE embeds coordination information directly into the carrier-frequency dynamics of an ongoing transmission, a receiver *need not demodulate or decode the primary payload to extract coordination information*. Instead, it must only synchronize to the signal sufficiently to estimate the CFO. This property is valuable in tactical settings where payload formats, encryption, or network credentials may differ across systems, but where basic physical-layer observability remains possible.

We envision FLE being used to exchange low-rate coordination information such as frequency plans, channel-occupancy summaries, and network configuration information, thereby enabling lightweight peer-to-peer spectrum awareness without requiring prior coordination, protocol interoperability, or dependence on vulnerable backhaul links. As with any coordination channel, deployment in adversarial environments requires measures such as message authentication and replay protection. These measures can be implemented above the FLE physical layer and are subjects for future work.

IV. Implementation of Real-Time System

To demonstrate the performance of the single-carrier FLE, we implement our real-world pipeline in GNU Radio Companion as shown in Fig. 3. The primary signal consisted of QPSK with the square-root raised-cosine (SRRC) pulse shape, unity DC gain, and 50% excess bandwidth. The secondary signal was a binary PAM pulse train using the SRRC pulse shape with 100% excess bandwidth. For the computer simulation, additive white Gaussian noise (AWGN) was added using GNU Radio's channel model block. For the primary channel, symbol timing synchronization was performed using the Gardner timing error detector [12] and a Farrow interpolator for timing adjustments. Carrier phase synchronization was performed using a second-order PLL with a closed-loop equivalent bandwidth $B_nT_{s,1} = 0.02$ cycles/symbol. Secondary channel detection was performed using the matched filter followed by a second symbol timing synchronizer using the Gardner timing error detector and Farrow interpolator. Experiments were conducted using a variety of values for $\Delta fT_{s,1}$ and N, as summarized in Section V.

With the same GNU Radio Companion environment used in simulation, we transmit our FLE overlay over the air and decode it in real time. In this setup, both the primary and FLE signals use a packetized structure with a 32-bit

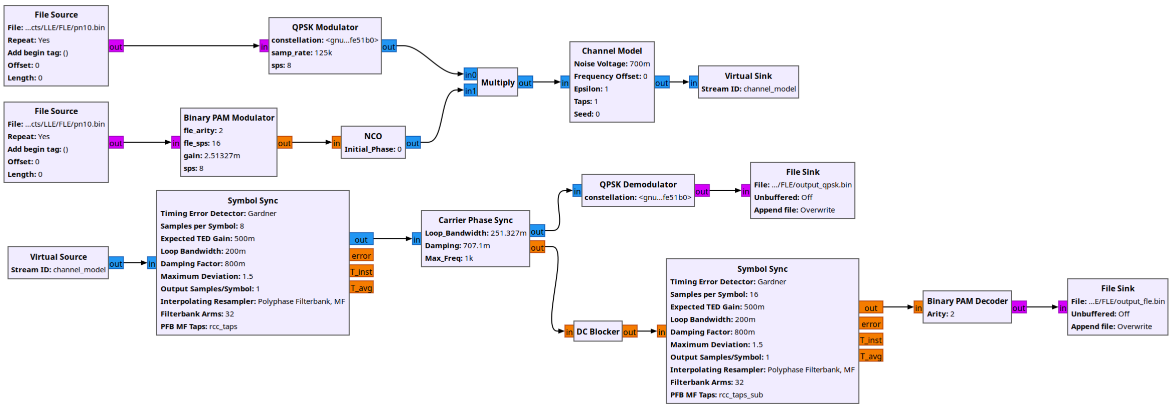

Fig. 3. The GNU Radio flow for FLE simulation

sync word marking the beginning of the packet and a 32-bit CRC to ensure signal integrity. We use two USRP B210 SDRs as the transmitter and receiver of the primary protocol. The transmitter is configured to concurrently transmit FLE data. We use the ADALM-Pluto SDR as a secondary RX, which passively monitors the transmission and extracts only the FLE information embedded in the frequency offsets. Using the relatively inexpensive Pluto SDR shows the low barrier to entry for our protocol and the universality of this design across different hardware capabilities. For our over-the-air experiments, we transmit in the unlicensed 915 MHz band. This real-time over-the-air setup validates the real-world capabilities of FLE to enable spectrum sharing and coordination.

V. Evaluation

In our evaluation, we seek to answer three main questions: (1) How much does FLE affect the performance of the primary communication link? (2) Under what conditions can the secondary channel be reliably decoded? (3) How do the FLE parameters $\Delta f T_{s,1}$ and N affect system performance? Performance is measured using bit-error rate (BER) as a function of E_b/N_0 for both the primary and secondary protocols, calculated using the bit energy of the primary protocol.

A. Simulation

After extensive simulation, we find that although FLE parameters can influence QPSK performance, their impact on the primary protocol remains minimal provided the parameters lie within a defined operating envelope. Fig. 4(a) shows the BER performance of QPSK under four different FLE configurations, along with a baseline case in which FLE is absent. For reference, the theoretical BER curve under AWGN is also included. The observed implementation loss is attributable to the simulation tooling rather than the presence of FLE. Nearly all simulated curves, including the no-FLE case, follow the same trajectory. Only when parameters are pushed to extreme values (e.g., $N = 16$ and $\Delta f T_{s,1} = 0.0064$) does a slight degradation appear. Even in this case, the degradation is limited to approximately 1 dB in E_b/N_0.

To identify the bounds of this stability envelope, we perform a parameter sweep, shown in Fig. 4(b). Primary-protocol reception remains near-perfect below a clear boundary in the N–$\Delta f T_{s,1}$ plane, indicating a wide operating region where FLE can coexist with QPSK without significant impact.

In contrast, the effect of FLE parameters on the subprotocol is more prominent. As shown in Fig. 4(c), both N and $\Delta f T_{s,1}$ strongly affect the BER performance of FLE. Increasing these parameters improves subprotocol detectability, but excessive values lead to degradation of the primary protocol, presenting a trade-off to the end user. Importantly, there exists an overlap region where both the primary protocol and the FLE subprotocol achieve high performance. This is evident from the FLE parameter sweep in Fig. 4(d).

We present the following as a design rule. We first take the intersection of the primary protocol and subprotocol stability regions and then apply a conservative linear fit to the boundary. This places the operating envelope at

$$0.0016 \leq \Delta f T_{s,1} \leq 0.0012N + 0.0080 \quad \text{for } N \geq 12 \quad (18)$$

Fig. 5 reveals what was predicted in the analysis of Section III-B. Here, we show the envelope described in (18) against the lock limits of (16) evaluated at the loop bandwidth used here. The measured boundary is linear as predicted and lies directly between the two bounds. The vertical edge at $N = 12$ is far more abrupt than the analysis predicts, because (16) places bounds based on the primary link integrity rather than the secondary symbol reception. Below this value the loop violates the bandwidth condition on (12) and smears the secondary pulse train. Empirically, this was found to be an abrupt edge, the subprotocol failing over a single step in N rather than degrading gradually.

Likewise, the horizontal edge of the envelope comes from the subprotocol SNR (15). Below $\Delta f T_{s,1} = 0.0016$ the deviation is too small to lift the secondary pulse train clear of the loop-filter noise, and the subprotocol bit-error rate collapses in Fig. 4(d) while the primary link remains untouched. This flat edge is itself predicted by (15): the ratio depends on $(\Delta f T_{s,1})^2$, on I_2, and on the primary E_{avg}/N_0, but not on N,

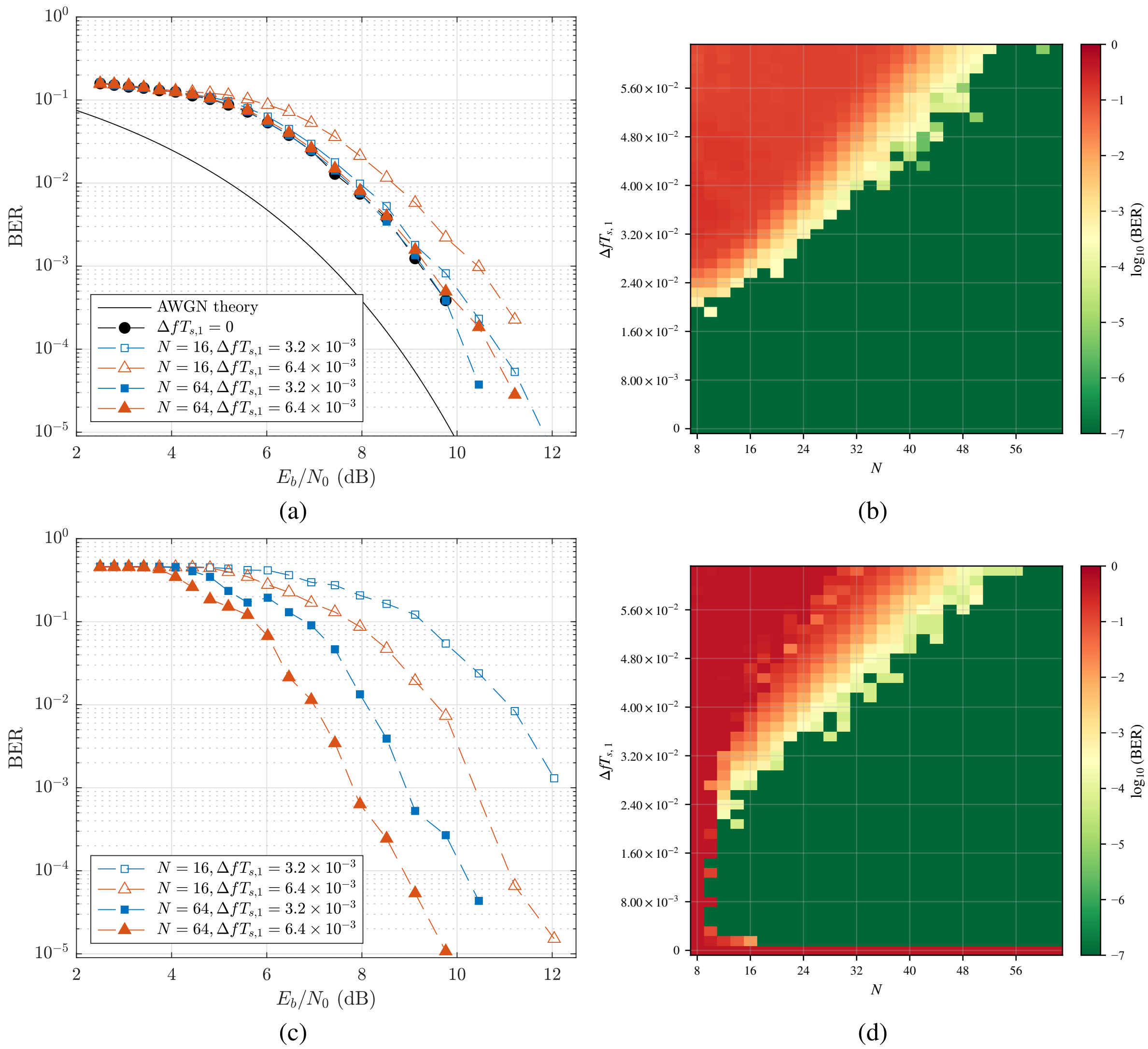

Fig. 4. Single-carrier QPSK performance under FLE: (a) primary channel BER for representative FLE parameter settings; (b) primary channel BER in the N–$\Delta f T_{s,1}$ parameter space for $E_b/N_0 = 18$ dB; (c) secondary channel BER for representative FLE parameter settings; (d) secondary channel BER in the N–$\Delta f T_{s,1}$ parameter space for $E_b/N_0 = 18$ dB.

so the detection floor is independent of the secondary symbol rate.

Using standard ITU channel models (EPA, EVA, EVB) [16], preliminary evaluations showed that more complex propagation environments—including frequency-selective fading and multipath effects—did not significantly degrade subprotocol performance. In several harsher channel conditions, the subprotocol even outperformed the underlying protocol. This behavior is attributed to the longer effective symbol duration, which improves robustness against multipath propagation, as well as the fact that the overlay shifts the entire base signal bandwidth uniformly, making localized frequency-selective impairments less pronounced. As a result, this work focuses primarily on the AWGN channel model. Mobility and oscillator drift were not explicitly tested in this work. As discussed in Section III-A, they enter the received signal only through $\vartheta'(t)$ and are expected to be benign except under high mobility combined with fast frequency-selective fading; a full tactical-channel characterization is left to future work.

B. Over-the-Air Tests

For the over-the-air experiments, operating parameters were selected based on the simulation results: a QPSK primary protocol with an FLE overlay using $N = 32$ and $\Delta f T_{s,1} = 0.0096$. Experiments were conducted at transmitter–receiver separations of 1 m and 20 m. These experiments serve as a preliminary feasibility study, not a full tactical-channel validation. At the shorter distance, near-perfect reception was observed for both the primary protocol and the FLE subprotocol. At the increased separation, the system continued to exhibit excellent performance, with both layers achieving measured BERs on the order of 10^{-6}.

Although the observed BERs were ultimately determined by the SNR conditions of the two scenarios, the key result is

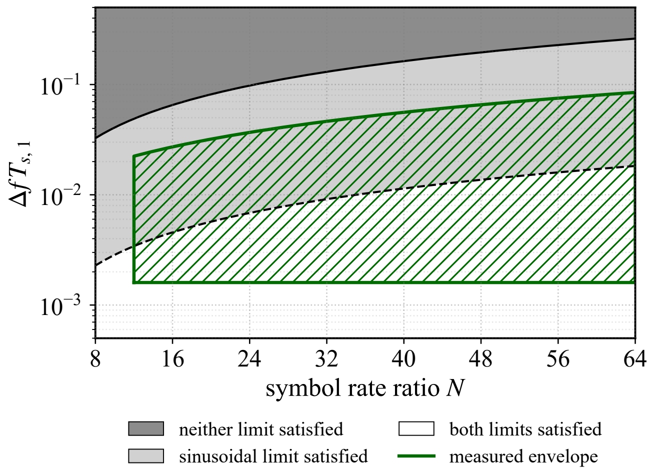

Fig. 5. The measured operating envelope compared with the lock limits (16), evaluated at the simulated loop bandwidth $B_n T_{s,1} = 0.02$. The measured envelope from simulation lies between the two limits as expected.

that the primary protocol and the subprotocol exhibited similar error-rate trends and operational ranges. This is precisely the behavior predicted by (15): the SNR at the secondary detector is proportional to the E_{avg}/N_0 of the primary link, so the two channels track one another as the link budget changes rather than failing at independent thresholds. These experiments demonstrate that the secondary signaling layer can closely track the reliability of the underlying communication system while remaining effectively transparent to the primary protocol. In tested configurations, the subprotocol achieved data rates exceeding 900 kbps, with performance ultimately limited by the computational capability of the laptops running GNU Radio Companion rather than by the FLE mechanism itself.

VI. Conclusion

This paper introduces FLE, a protocol-agnostic secondary signaling technique that can be used to embed spectrum coordination information into an existing wireless transmission through small, controlled frequency offsets.

We analyzed FLE for a single-carrier system, deriving the SNR available to the secondary detector and closed-form lock limits that bound the secondary data rate against the impact on the primary protocol. The simulated operating envelope takes the form those limits predict and lies within the region they guarantee, and the over-the-air results show that primary and subprotocol communication in a real-world setting are feasible. These results show that excess synchronization margin can be repurposed as a lightweight, ad hoc signaling channel, enabling spectrum coordination between unassociated devices without requiring protocol interoperability, network association, or centralized infrastructure—properties well-aligned with the demands of tactical and coalition operations.

Future work will explore additional applications of the proposed secondary signaling layer, including control-plane coordination in highly congested networks, and a quantitative comparison against alternative uses of the same synchronization margin. Further investigation will also extend this approach and analysis to other intentional signal impairment domains, such as timing and phase, to assess their feasibility and impact on both primary and secondary communication performance.

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